Lecture 12: Mesh Current Analysis

EE201 – Circuit Theory I

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current, mesh current 1 voltage supermesh

Review: node voltage analysis

The procedure

- Ground; node voltages $\mathbf{e}$
- KVL automatic: $\mathbf{v} = A^T \mathbf{e}$
- KCL at non-ground nodes
- $Y_n \mathbf{e} = \mathbf{i}_n$

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Voltage sources

- To the ground: e_k known
- Between two nodes: supernode
- Dependent: one more constraint

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The opposite?

Motivation

Node voltages

- Unknowns: $n - 1$ node voltages
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- Unknowns: $\ell = b - n + 1$ mesh currents

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Motivation

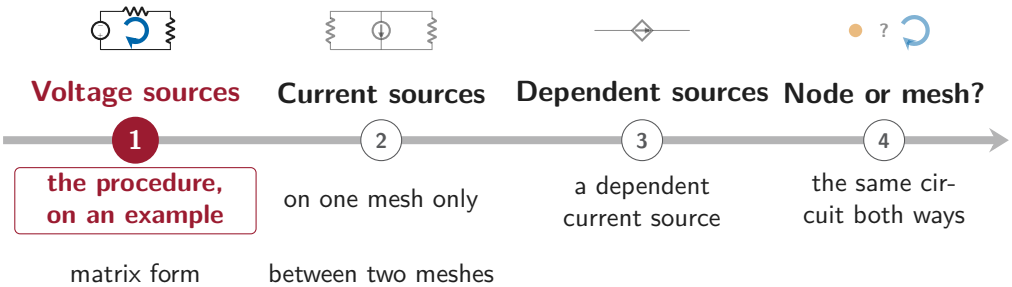
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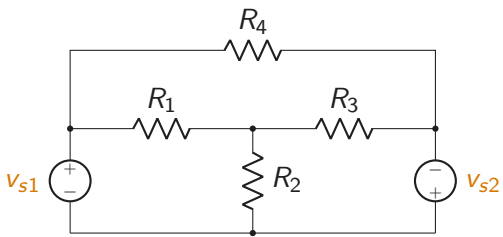
- Unknowns: $\ell = b - n + 1$ mesh currents
- KCL automatic: $\mathbf{i} = M^T \mathbf{i}_M$
- Write KVL
- Needs a planar circuit

Roadmap


Voltage sources
Current sources
Dependent sources
Node or mesh?
1
2
3
4
the procedure, on an example
matrix form
on one mesh only
between two meshes
a dependent current source
the same circuit both ways

Example 12.1: mesh current analysis step by step

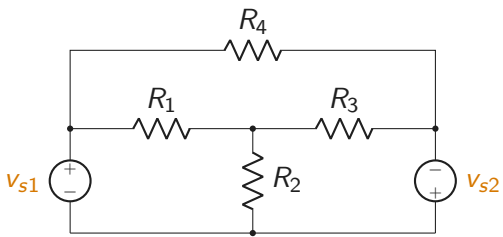
For now: only resistors and independent voltage sources



Question. How many unknowns?

Example 12.1: mesh current analysis step by step

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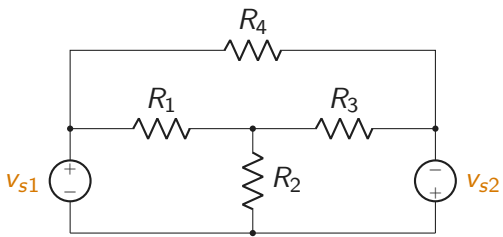


Question. How many unknowns?

$b = 6$: 12 unknowns

Example 12.1: mesh current analysis step by step

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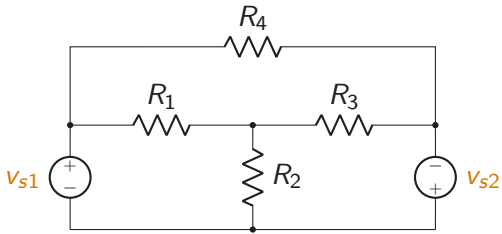


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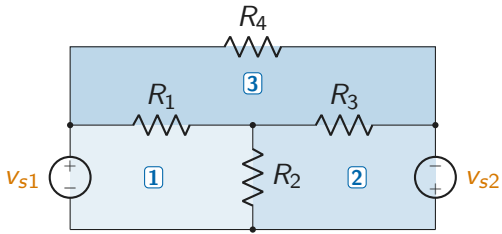
Mesh current analysis: 3

Step 1: find the inner meshes



Planar: no crossing branches

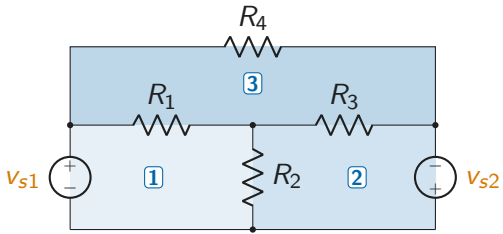
Step 1: find the inner meshes



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Inner meshes: the windows

Step 1: find the inner meshes

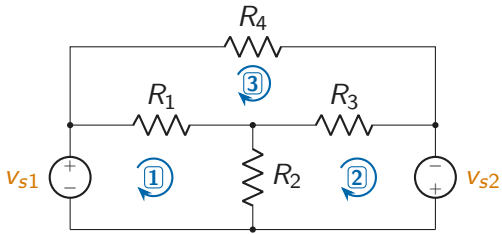


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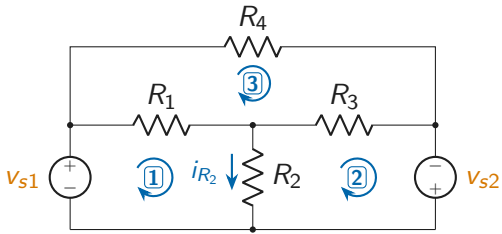
$$\ell = b - n + 1 = 6 - 4 + 1 = 3$$

Step 2: assign the mesh currents



Unknowns, clockwise: i_{M1}, i_{M2}, i_{M3}

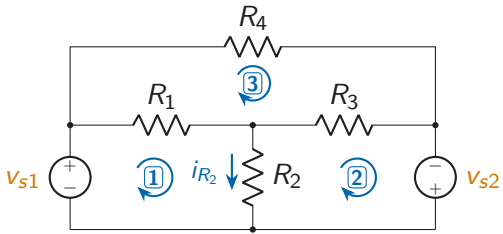
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$i_{R_2} = i_{M1} - i_{M2}$

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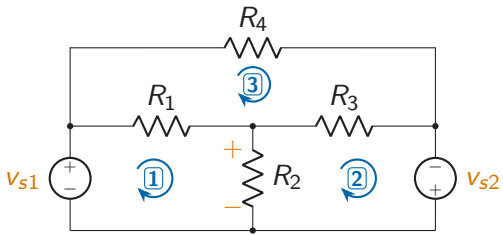


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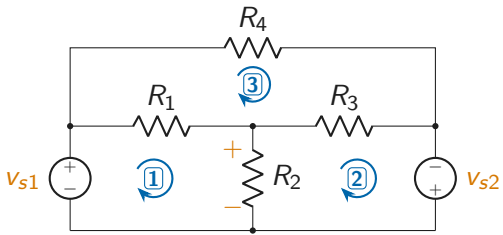
KCL holds automatically

Step 3: write the resistor voltages with mesh currents



Question. v of R_2 along mesh 1?

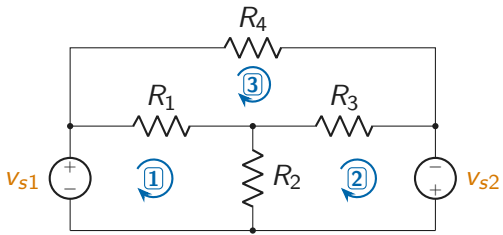
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Question. v of R_2 along mesh 1?

$v = R_2(i_{M1} - i_{M2})$

Step 3: write the resistor voltages with mesh currents

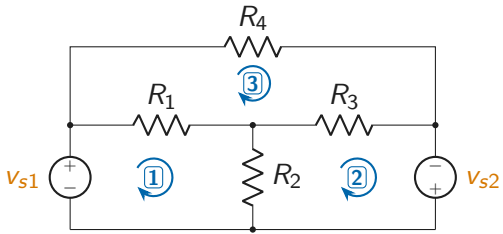


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R_4: $v = R_4\, i_{M3}$

Step 3: write the resistor voltages with mesh currents



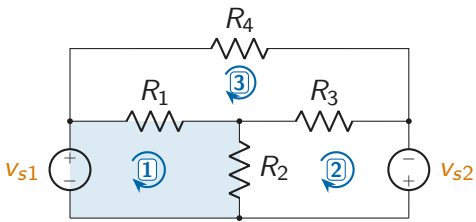
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R (own $-$ other)

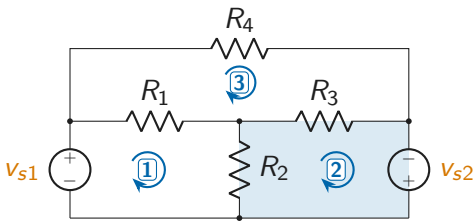
Step 4: KVL around every inner mesh



Mesh 1:

$$-v_{s1} + R_1(i_{M1} - i_{M3}) + R_2(i_{M1} - i_{M2}) = 0$$

Step 4: KVL around every inner mesh



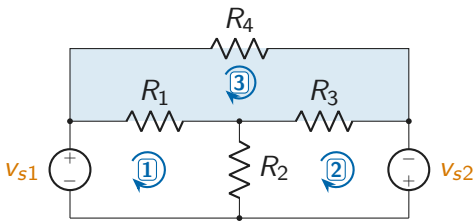
Mesh 1:

$$-v_{s1} + R_1(i_{M1} - i_{M3}) + R_2(i_{M1} - i_{M2}) = 0$$

Mesh 2:

$$R_2(i_{M2} - i_{M1}) + R_3(i_{M2} - i_{M3}) - v_{s2} = 0$$

Step 4: KVL around every inner mesh



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$$-v_{s1} + R_1(i_{M1} - i_{M3}) + R_2(i_{M1} - i_{M2}) = 0$$

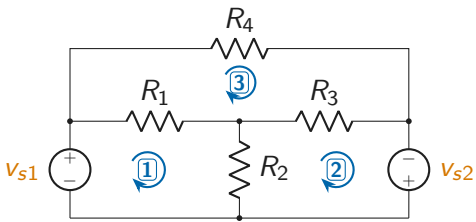
Mesh 2:

$$R_2(i_{M2} - i_{M1}) + R_3(i_{M2} - i_{M3}) - v_{s2} = 0$$

Mesh 3:

$$R_4\, i_{M3} + R_3(i_{M3} - i_{M2}) + R_1(i_{M3} - i_{M1}) = 0$$

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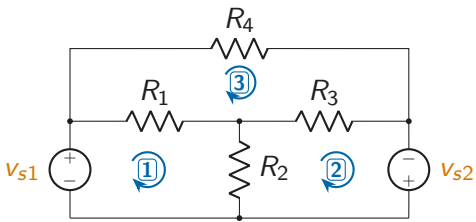
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Source met $-$ to $+$: minus

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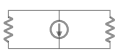
$$R_4\, i_{M3} + R_3(i_{M3} - i_{M2}) + R_1(i_{M3} - i_{M1}) = 0$$

Source met $-$ to $+$: minus

3 equations, 3 unknowns

Roadmap









Voltage sources | **Current sources** | **Dependent sources** | **Node or mesh?**

1 — the procedure, on an example ✓

2 — on one mesh only

3 — a dependent current source

4 — the same circuit both ways

matrix form between two meshes

Example 12.1 in matrix form

$$
\begin{aligned}
(R_1 + R_2)\, i_{M1} - R_2\, i_{M2} - R_1\, i_{M3} &= v_{s1} \\
-R_2\, i_{M1} + (R_2 + R_3)\, i_{M2} - R_3\, i_{M3} &= v_{s2} \\
-R_1\, i_{M1} - R_3\, i_{M2} + (R_1 + R_3 + R_4)\, i_{M3} &= 0
\end{aligned}
$$

Question. One matrix equation?

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Z_M: mesh impedance matrix

Example 12.1 in matrix form

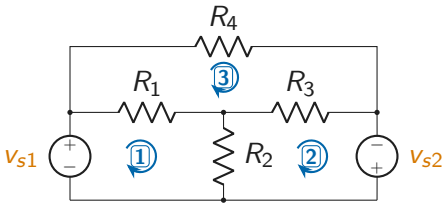
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$$

Z_M: mesh impedance matrix

$\mathbf{v}_M$: source voltages rising along each mesh

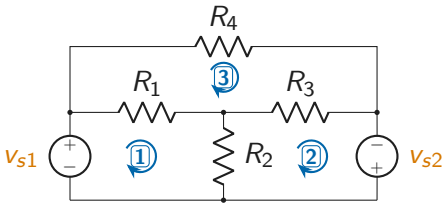
Writing Z_M directly from the circuit



Question. Read Z_M from the circuit?

$$\underbrace{\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} \\ \\ \end{bmatrix}}_{\mathbf{v}_M}$$

Writing Z_M directly from the circuit

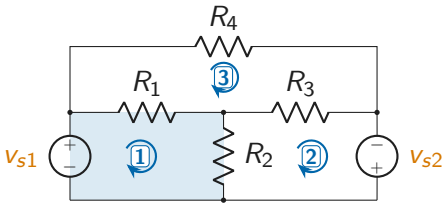


Question. Read Z_M from the circuit?

Diagonal (k, k): $+\sum R$ around mesh k

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Writing Z_M directly from the circuit

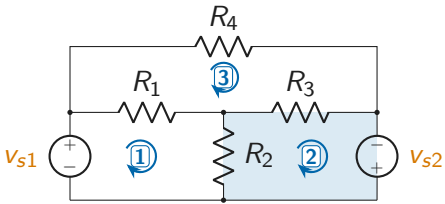


Question. Read Z_M from the circuit?

Diagonal (k, k): $+\sum R$ around mesh k

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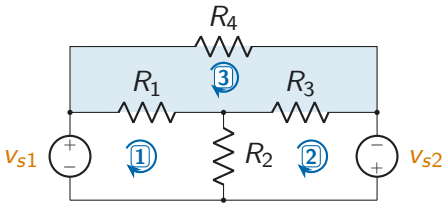


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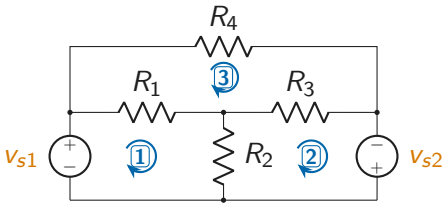


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Writing Z_M directly from the circuit



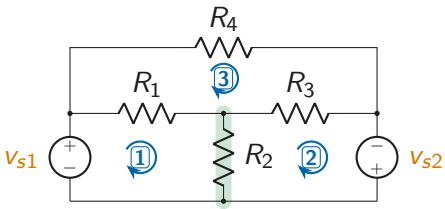
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Diagonal (k, k): $+\sum R$ around mesh k

Off-diagonal (k, j): $-\sum R$ shared by k, j

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Writing Z_M directly from the circuit



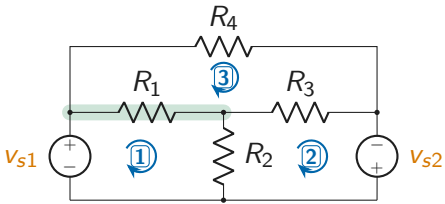
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$$\underbrace{\begin{bmatrix} R_1 + R_2 & -R_2 & \\ -R_2 & R_2 + R_3 & \\ & & R_1 + R_3 + R_4 \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} \\ \\ \end{bmatrix}}_{\mathbf{v}_M}$$

Writing Z_M directly from the circuit



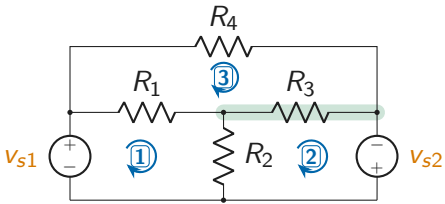
Question. Read Z_M from the circuit?

Diagonal (k,k): $+\sum R$ around mesh k

Off-diagonal (k,j): $-\sum R$ shared by k, j

$$\underbrace{\begin{bmatrix} R_1+R_2 & -R_2 & -R_1 \\ -R_2 & R_2+R_3 & \\ -R_1 & & R_1+R_3+R_4 \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} \\ \\ \end{bmatrix}}_{\mathbf{v}_M}$$

Writing Z_M directly from the circuit



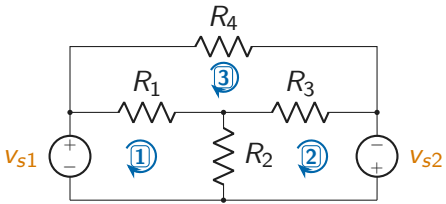
Question. Read Z_M from the circuit?

Diagonal (k,k): $+\sum R$ around mesh k

Off-diagonal (k,j): $-\sum R$ shared by k, j

$$\underbrace{\begin{bmatrix} R_1+R_2 & -R_2 & -R_1 \\ -R_2 & R_2+R_3 & -R_3 \\ -R_1 & -R_3 & R_1+R_3+R_4 \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} \\ \\ \end{bmatrix}}_{\mathbf{v}_M}$$

Writing Z_M directly from the circuit



Question. Read Z_M from the circuit?

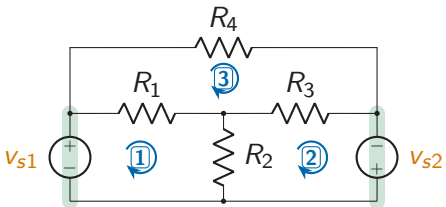
Diagonal (k,k): $+\sum R$ around mesh k

Off-diagonal (k,j): $-\sum R$ shared by k, j

Right side k: $\sum v_s$ rising along mesh k

$$\underbrace{\begin{bmatrix} R_1+R_2 & -R_2 & -R_1 \\ -R_2 & R_2+R_3 & -R_3 \\ -R_1 & -R_3 & R_1+R_3+R_4 \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} \\ \\ \\ \end{bmatrix}}_{\mathbf{v}_M}$$

Writing Z_M directly from the circuit



Question. Read Z_M from the circuit?

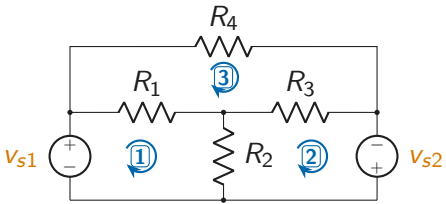
Diagonal (k,k): $+\sum R$ around mesh k

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Right side k: $\sum v_s$ rising along mesh k

$$\underbrace{\begin{bmatrix} R_1+R_2 & -R_2 & -R_1 \\ -R_2 & R_2+R_3 & -R_3 \\ -R_1 & -R_3 & R_1+R_3+R_4 \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix}}_{\mathbf{v}_M}$$

Writing Z_M directly from the circuit



Question. Read Z_M from the circuit?

Diagonal (k,k): $+\sum R$ around mesh k

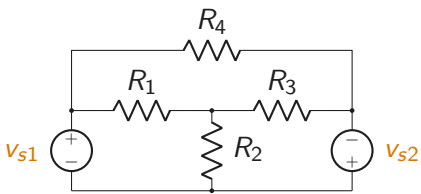
Off-diagonal (k,j): $-\sum R$ shared by k, j

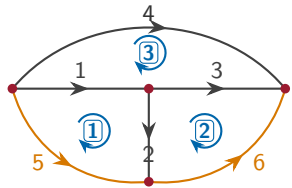
Right side k: $\sum v_s$ rising along mesh k

$$\underbrace{\begin{bmatrix} R_1+R_2 & -R_2 & -R_1 \\ -R_2 & R_2+R_3 & -R_3 \\ -R_1 & -R_3 & R_1+R_3+R_4 \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix}}_{\mathbf{v}_M}$$

Same as KVL ✓ Z_M is symmetric

Z_M from the graph: the mesh matrix

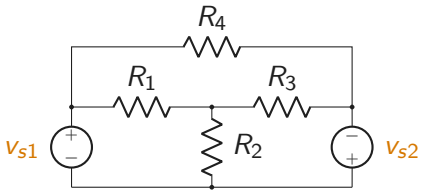


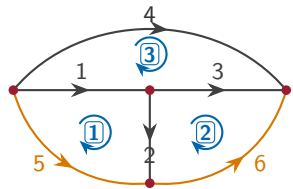


1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2} ($+$ to $-$)

Question. Mesh matrix M?

Z_M from the graph: the mesh matrix





1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2} ($+$ to $-$)

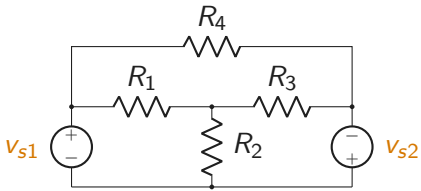
Question. Mesh matrix M?

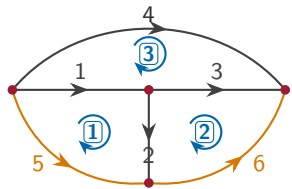
One row per inner mesh

$+1$ along, -1 against

$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \left[\begin{array}{cccccc} & & & & & \\ & & & & & \\ & & & & & \end{array}\right] \end{array}$$

Z_M from the graph: the mesh matrix





1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2} (+ to −)

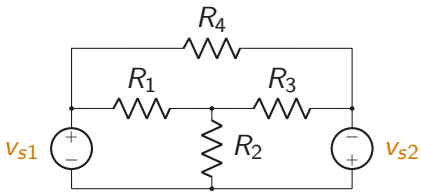
Question. Mesh matrix M?

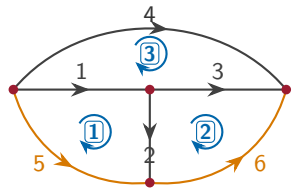
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$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & -1 & 0 \\ & & & & & \\ & & & & & \end{bmatrix} \end{array}$$

Z_M from the graph: the mesh matrix





1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2} (+ to −)

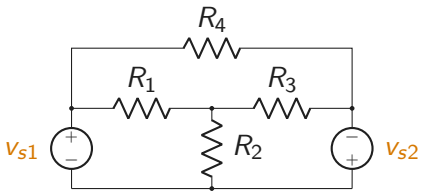
Question. Mesh matrix M?

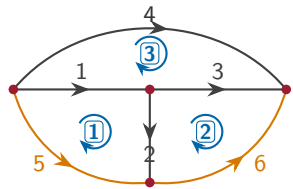
One row per inner mesh

$+1$ along, -1 against

$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 1 & 0 & 0 & -1 \\ & & & & & \end{bmatrix} \end{array}$$

Z_M from the graph: the mesh matrix





1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2} (+ to −)

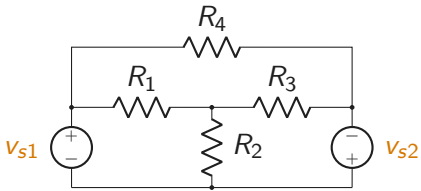
Question. Mesh matrix M?

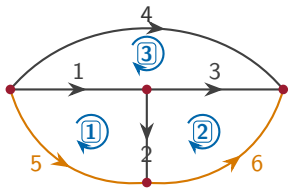
One row per inner mesh

$+1$ along, -1 against

$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 1 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \end{bmatrix} \end{array}$$

Z_M from the graph: the mesh matrix





1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2} (+ to −)

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One row per inner mesh

$+1$ along, -1 against

$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \left[\begin{array}{cccc|cc} 1 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 1 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \end{array}\right] \end{array}$$

M_R: resistors M_S: sources

Z_M from the graph: KVL, KCL and Ohm's law in blocks

Resistors first, then voltage sources:

$$M = [\, M_R \;\; M_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}$$

Z_M from the graph: KVL, KCL and Ohm's law in blocks

Resistors first, then voltage sources:

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KVL: $M\mathbf{v} = [\, M_R \;\; M_S \,] \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix} = M_R\mathbf{v}_R + M_S\mathbf{v}_S = \mathbf{0}$

Z_M from the graph: KVL, KCL and Ohm's law in blocks

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KVL: $M\mathbf{v} = [\, M_R \;\; M_S \,] \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix} = M_R\mathbf{v}_R + M_S\mathbf{v}_S = \mathbf{0}$

KCL:

$$\begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix} = \begin{bmatrix} M_R^T \\ M_S^T \end{bmatrix} \mathbf{i}_M$$

Z_M from the graph: KVL, KCL and Ohm's law in blocks

Resistors first, then voltage sources:

$$M = [\, M_R \;\; M_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}$$

KVL: $M\mathbf{v} = [\, M_R \;\; M_S \,] \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix} = M_R\mathbf{v}_R + M_S\mathbf{v}_S = \mathbf{0}$

KCL:

$$\begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix} = \begin{bmatrix} M_R^T \\ M_S^T \end{bmatrix} \mathbf{i}_M \qquad \Rightarrow \qquad \mathbf{i}_R = M_R^T \mathbf{i}_M \;\; \text{(resistor rows only)}$$

Z_M from the graph: KVL, KCL and Ohm's law in blocks

Resistors first, then voltage sources:

$$M = [\, M_R \;\; M_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}$$

KVL: $M\mathbf{v} = [\, M_R \;\; M_S \,] \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix} = M_R\mathbf{v}_R + M_S\mathbf{v}_S = \mathbf{0}$

KCL:

$$\begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix} = \begin{bmatrix} M_R^T \\ M_S^T \end{bmatrix} \mathbf{i}_M \qquad \Rightarrow \qquad \mathbf{i}_R = M_R^T \mathbf{i}_M \;\; \text{(resistor rows only)}$$

Ohm: $\mathbf{v}_R = R\,\mathbf{i}_R, \quad R = \mathrm{diag}(R_1, \ldots, R_{b_R})$

Z_M from the graph: the general result

Ohm and KCL into KVL:

$$M_R \mathbf{v}_R + M_S \mathbf{v}_S = M_R\, R\, \underbrace{M_R^T \mathbf{i}_M}_{\mathbf{i}_R} + M_S \mathbf{v}_S = \mathbf{0} \quad \Rightarrow \quad M_R\, R\, M_R^T\, \mathbf{i}_M = -M_S \mathbf{v}_S$$

Z_M from the graph: the general result

Ohm and KCL into KVL:

$$M_R \mathbf{v}_R + M_S \mathbf{v}_S = M_R\, R\, \underbrace{M_R^T \mathbf{i}_M}_{\mathbf{i}_R} + M_S \mathbf{v}_S = \mathbf{0} \quad \Rightarrow \quad M_R\, R\, M_R^T\, \mathbf{i}_M = -M_S \mathbf{v}_S$$

$$Z_M = M_R\, R\, M_R^T, \qquad \mathbf{v}_M = -M_S \mathbf{v}_S$$

Z_M from the graph: the general result

Ohm and KCL into KVL:

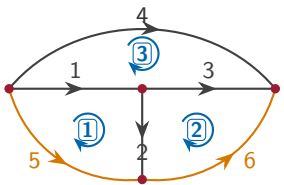
$$M_R \mathbf{v}_R + M_S \mathbf{v}_S = M_R\, R \underbrace{M_R^T \mathbf{i}_M}_{\mathbf{i}_R} + M_S \mathbf{v}_S = \mathbf{0} \quad \Rightarrow \quad M_R\, R\, M_R^T\, \mathbf{i}_M = -M_S \mathbf{v}_S$$

$$Z_M = M_R\, R\, M_R^T, \qquad \mathbf{v}_M = -M_S \mathbf{v}_S$$

$$Z_M^T = M_R\, R^T M_R^T = Z_M: \quad \text{always symmetric}$$

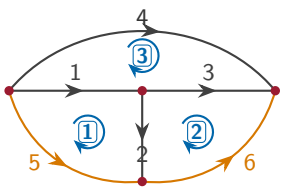
Z_M from the graph: Example 12.1, the resistor currents

M_R^T: columns 1–4 of M, as rows



Z_M from the graph: Example 12.1, the resistor currents

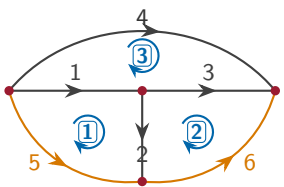
M_R^T: columns 1–4 of M, as rows



$$\underbrace{\begin{bmatrix} i_{R_1} \\ i_{R_2} \\ i_{R_3} \\ i_{R_4} \end{bmatrix}}_{\mathbf{i}_R} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{M_R^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix}$$

Z_M from the graph: Example 12.1, the resistor currents

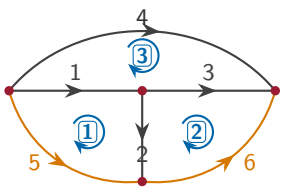
M_R^T: columns 1–4 of M, as rows



$$\underbrace{\begin{bmatrix} i_{R_1} \\ i_{R_2} \\ i_{R_3} \\ i_{R_4} \end{bmatrix}}_{\mathbf{i}_R} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{M_R^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} i_{M1} - i_{M3} \\ i_{M1} - i_{M2} \\ i_{M2} - i_{M3} \\ i_{M3} \end{bmatrix}$$

Z_M from the graph: Example 12.1, the resistor currents

M_R^T: columns 1–4 of M, as rows

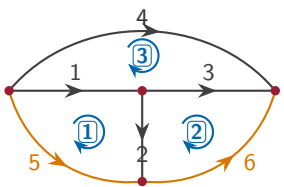


$$\underbrace{\begin{bmatrix} i_{R_1} \\ i_{R_2} \\ i_{R_3} \\ i_{R_4} \end{bmatrix}}_{\mathbf{i}_R} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{M_R^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} i_{M1} - i_{M3} \\ i_{M1} - i_{M2} \\ i_{M2} - i_{M3} \\ i_{M3} \end{bmatrix}$$

Along minus against

Z_M from the graph: Example 12.1, the resistor currents

M_R^T: columns 1–4 of M, as rows



$$\underbrace{\begin{bmatrix} i_{R_1} \\ i_{R_2} \\ i_{R_3} \\ i_{R_4} \end{bmatrix}}_{\mathbf{i}_R} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{M_R^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} i_{M1} - i_{M3} \\ i_{M1} - i_{M2} \\ i_{M2} - i_{M3} \\ i_{M3} \end{bmatrix}$$

Along minus against

Ohm: $v_{R_k} = R_k i_{R_k}$

Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix}$$

Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix} = Z_M \quad \checkmark$$

Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix} = Z_M \quad \checkmark$$

$$-M_S \mathbf{v}_S = - \begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_{s1} \\ v_{s2} \end{bmatrix} = \begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix}$$

Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix} = Z_M \quad \checkmark$$

$$-M_S \mathbf{v}_S = -\begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_{s1} \\ v_{s2} \end{bmatrix} = \begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix} = \mathbf{v}_M \quad \checkmark$$

Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

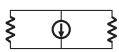
$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix} = Z_M \quad \checkmark$$

$$-M_S \mathbf{v}_S = -\begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_{s1} \\ v_{s2} \end{bmatrix} = \begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix} = \mathbf{v}_M \quad \checkmark$$

The rules = this product

Roadmap









Voltage sources

Current sources

Dependent sources

Node or mesh?

✓

2

3

4

the procedure, on an example ✓

matrix form ✓

on one mesh only

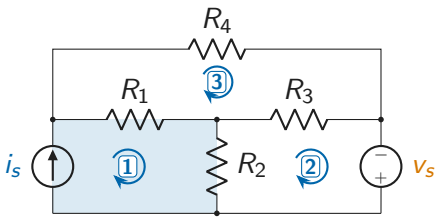
between two meshes

a dependent current source

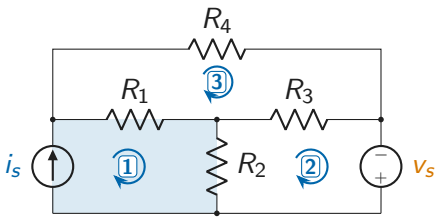
the same circuit both ways

Example 12.2: a current source on one mesh only

$i_{M1} = i_s$: known



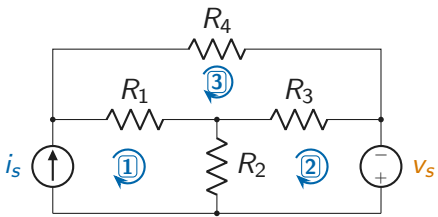
Example 12.2: a current source on one mesh only



$i_{M1} = i_s$: known

No KVL around mesh 1. Unknowns: i_{M2}, i_{M3}

Example 12.2: a current source on one mesh only

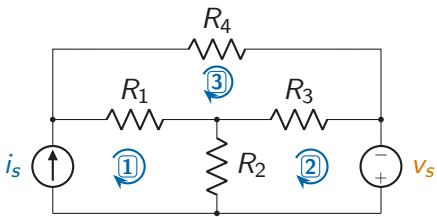


$i_{M1} = i_s$: known

No KVL around mesh 1. Unknowns: i_{M2}, i_{M3}

Mesh 2: $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

Example 12.2: a current source on one mesh only



$i_{M1} = i_s$: known

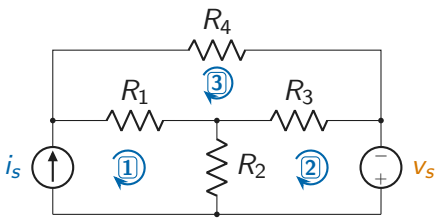
No KVL around mesh 1. Unknowns: i_{M2}, i_{M3}

Mesh 2: $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

Mesh 3:

$R_1(i_{M3} - i_s) + R_3(i_{M3} - i_{M2}) + R_4\, i_{M3} = 0$

Example 12.2: a current source on one mesh only



$i_{M1} = i_s$: known

No KVL around mesh 1. Unknowns: i_{M2}, i_{M3}

Mesh 2: $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

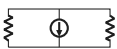
Mesh 3:

$R_1(i_{M3} - i_s) + R_3(i_{M3} - i_{M2}) + R_4\, i_{M3} = 0$

$$\begin{bmatrix} R_2 + R_3 & -R_3 \\ -R_3 & R_1 + R_3 + R_4 \end{bmatrix} \begin{bmatrix} i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} v_s + R_2 i_s \\ R_1 i_s \end{bmatrix}$$

Roadmap









Voltage sources ✓

the procedure, on an example ✓

matrix form ✓

Current sources 2

on one mesh only ✓

between two meshes

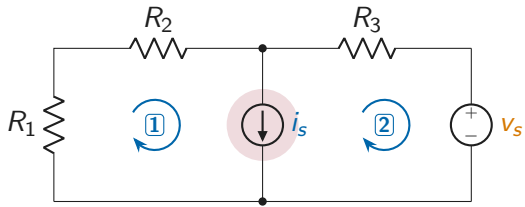
Dependent sources 3

a dependent current source

Node or mesh? 4

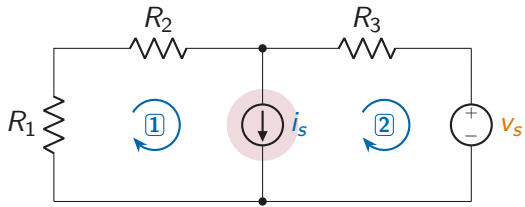
the same circuit both ways

Example 12.3: a current source between two meshes



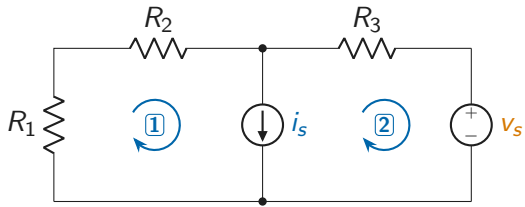
- **Question.** KVL around mesh 1: voltage across i_s?

Example 12.3: a current source between two meshes



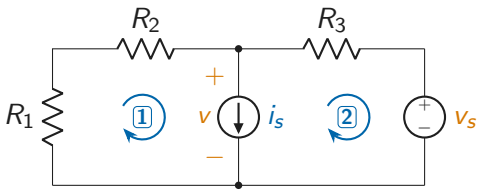
- **Question.** KVL around mesh 1: voltage across i_s?
- Not given by i_{M1}, i_{M2}: a current source fixes i, not v

Example 12.3: a current source between two meshes



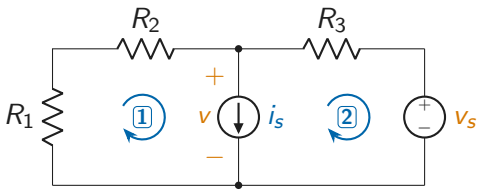
- **Question.** KVL around mesh 1: voltage across i_s?
- Not given by i_{M1}, i_{M2}: a current source fixes i, not v
- Method I: its voltage as an unknown Method II: supermesh

Method I: the voltage as an unknown



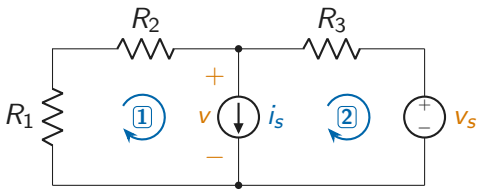
New unknown: v

Method I: the voltage as an unknown



Mesh 1: $(R_1 + R_2)\, i_{M1} + v = 0$ New unknown: v

Method I: the voltage as an unknown

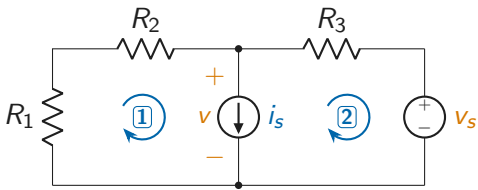


Mesh 1: $(R_1 + R_2)\, i_{M1} + v = 0$

New unknown: v

Mesh 2: $R_3\, i_{M2} + v_s - v = 0$

Method I: the voltage as an unknown



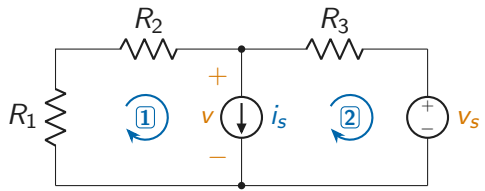
Mesh 1: $(R_1 + R_2)\, i_{M1} + v = 0$

New unknown: v

Mesh 2: $R_3\, i_{M2} + v_s - v = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

Method I: the voltage as an unknown

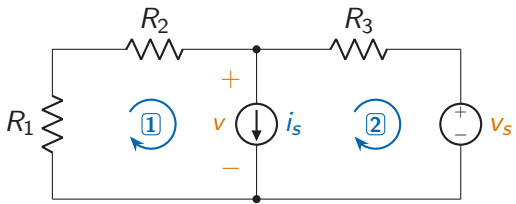


$$\begin{bmatrix} R_1+R_2 & 0 & 1 \\ 0 & R_3 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ -v_s \\ i_s \end{bmatrix}$$

New unknown: v

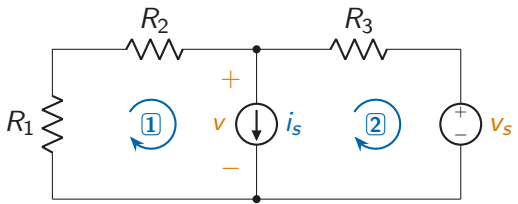
3 unknowns, 3 equations

Method II: the supermesh



Question. Can we get rid of v?

Method II: the supermesh

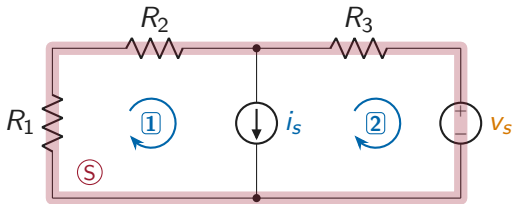


Question. Can we get rid of v?

Add KVL of meshes 1 and 2:

v cancels

Method II: the supermesh

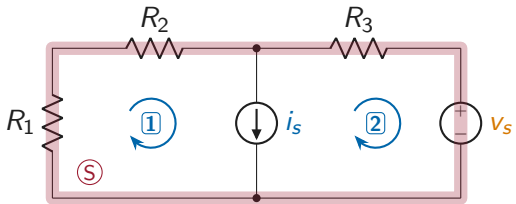


Question. Can we get rid of v? = KVL around the outer loop (S)

Add KVL of meshes 1 and 2:

v cancels

Method II: the supermesh



Question. Can we get rid of v?

Add KVL of meshes 1 and 2:

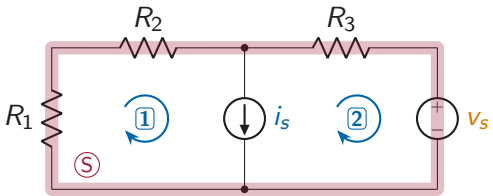
v cancels

$=$ KVL around the outer loop (S)

Supermesh

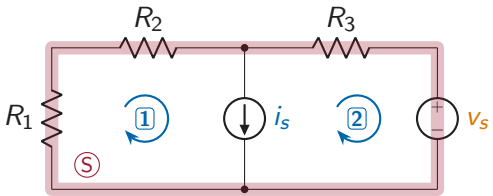
Two meshes as one, skipping the current source

Method II: the equations



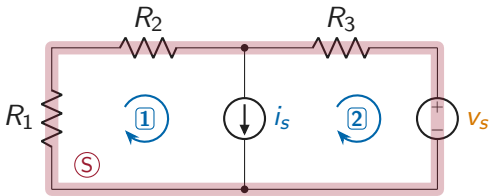
Question. KVL around the supermesh?

Method II: the equations



Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

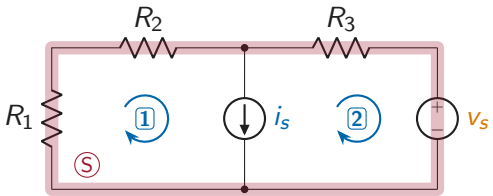
Method II: the equations



Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

Method II: the equations

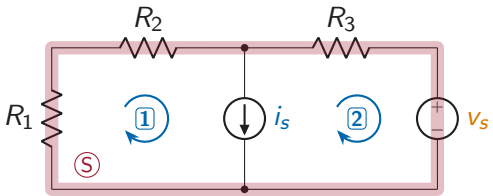


Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

$$\begin{bmatrix} R_1 + R_2 & R_3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} -v_s \\ i_s \end{bmatrix}$$

Method II: the equations



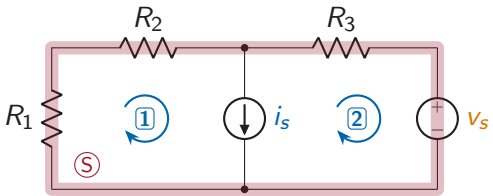
Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

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$$\begin{bmatrix} R_1 + R_2 & R_3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} -v_s \\ i_s \end{bmatrix}$$

2 unknowns (one fewer)

Method II: the equations



Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

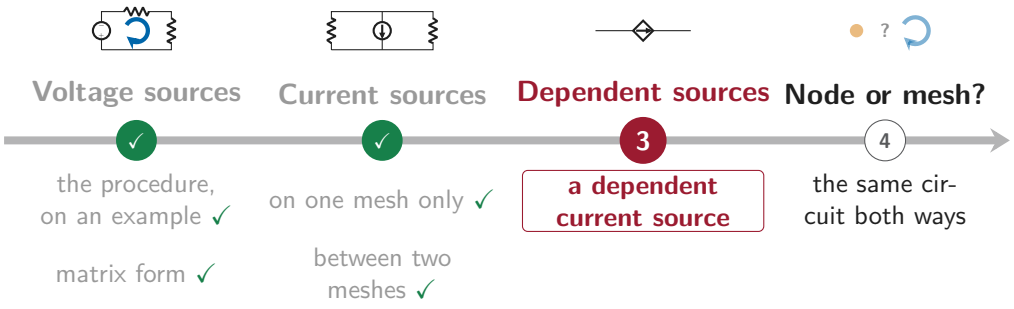
Constraint: $i_{M1} - i_{M2} = i_s$

$$\begin{bmatrix} R_1 + R_2 & R_3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} -v_s \\ i_s \end{bmatrix}$$

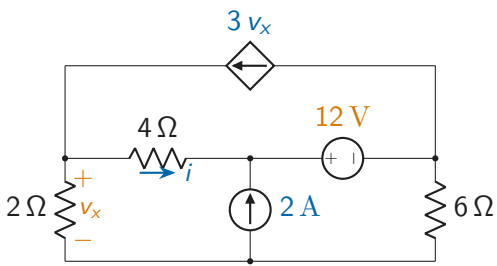
2 unknowns (one fewer)

Not symmetric

Roadmap

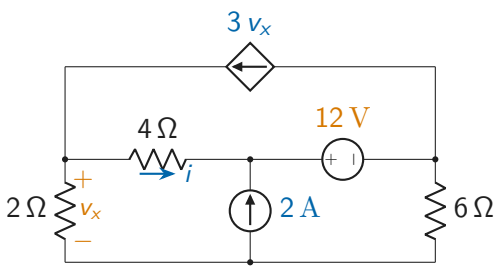

Voltage sources
Current sources
Dependent sources
Node or mesh?
✓
✓
3
4
the procedure, on an example ✓
matrix form ✓
on one mesh only ✓
between two meshes ✓
a dependent current source
the same circuit both ways

Example 12.4: a dependent current source



Question. Find i of the 4 Ω.

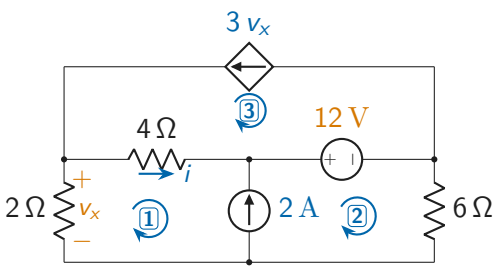
Example 12.4: a dependent current source



Question. Find i of the 4 Ω.

Treat $3\,v_x$ as a current source

Example 12.4: a dependent current source

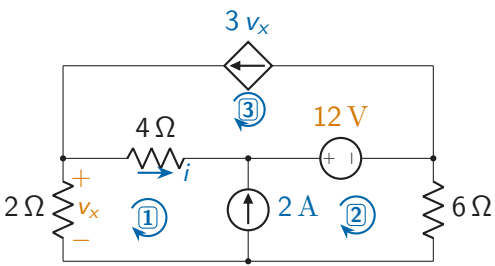


Question. Find i of the 4 Ω.

Treat $3\,v_x$ as a current source

Unknowns: i_{M1}, i_{M2}, i_{M3}

Example 12.4: a dependent current source



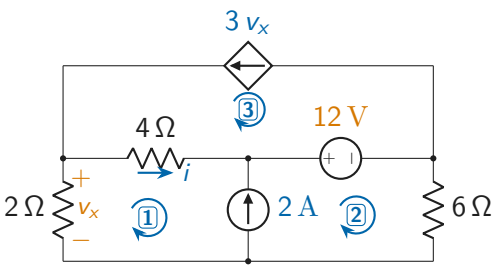
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Unknowns: i_{M1}, i_{M2}, i_{M3}

2 A between meshes 1, 2: supermesh

Example 12.4: a dependent current source



Question. Find i of the 4 Ω.

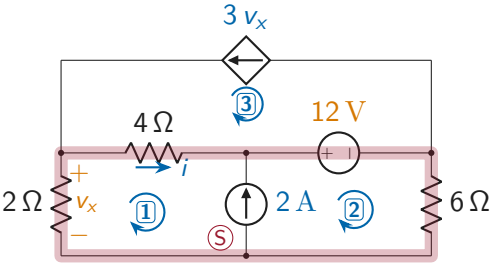
Treat $3\,v_x$ as a current source

Unknowns: i_{M1}, i_{M2}, i_{M3}

2 A between meshes 1, 2: supermesh

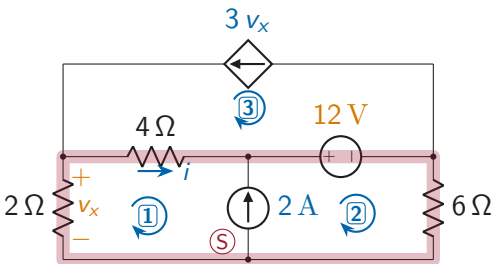
$3\,v_x$ on mesh 3 only: fixes i_{M3}

Example 12.4: the equations



Question. KVL around the supermesh?

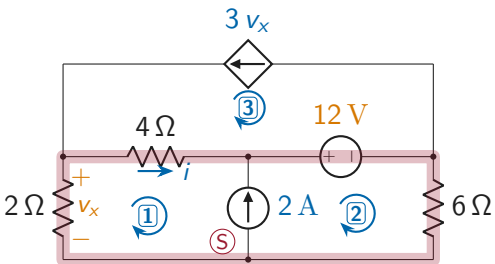
Example 12.4: the equations



Supermesh:

$$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$$

Example 12.4: the equations

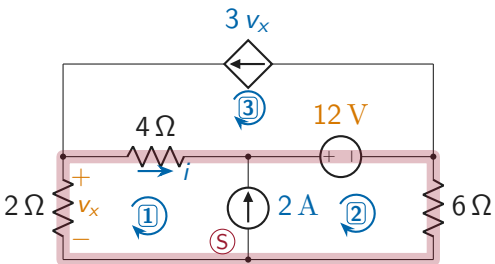


Supermesh:

$$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$$

2 A **source:** $i_{M2} - i_{M1} = 2 \quad (2)$

Example 12.4: the equations



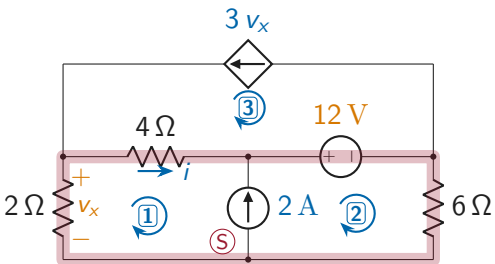
Supermesh:

$$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$$

2 A **source:** $i_{M2} - i_{M1} = 2 \quad (2)$

Dependent source: $i_{M3} = -3\,v_x$

Example 12.4: the equations



Supermesh:

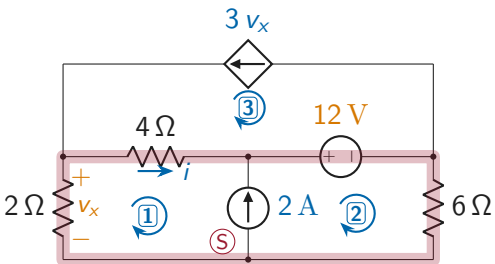
$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$

$2\,\mathrm{A}$ **source:** $i_{M2} - i_{M1} = 2 \quad (2)$

Dependent source: $i_{M3} = -3\,v_x$

Question. v_x in mesh currents?

Example 12.4: the equations



Supermesh:

$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$

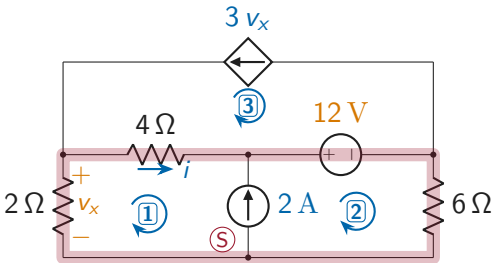
$2\,\mathrm{A}$ **source:** $i_{M2} - i_{M1} = 2 \quad (2)$

Dependent source: $i_{M3} = -3\,v_x$

Question. v_x in mesh currents?

$v_x = -2\,i_{M1}$

Example 12.4: the equations



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Constraint: $i_{M3} = 6\,i_{M1} \quad (3)$

Example 12.4: matrix form

(1), (2), (3):

$$\begin{bmatrix} 6 & 6 & -4 \\ -1 & 1 & 0 \\ -6 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} -12 \\ 2 \\ 0 \end{bmatrix}$$

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$$i = i_{M1} - i_{M3} = 2 - 12 = -10\,\mathrm{A}$$

Dependent sources

- Like an independent source of the same kind

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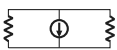
Remark

Dependent current source on mesh k only:

no KVL around mesh k; write the constraint instead

Roadmap









Voltage sources ✓

the procedure, on an example ✓

matrix form ✓

Current sources ✓

on one mesh only ✓

between two meshes ✓

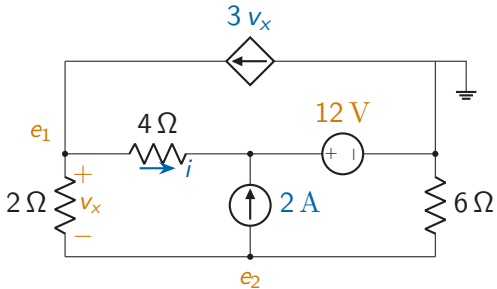
Dependent sources ✓

a dependent current source ✓

Node or mesh? 4

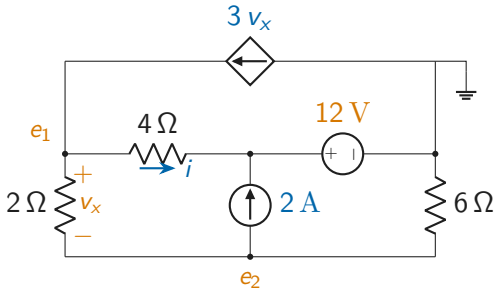
the same circuit both ways

Example 12.4 with node voltages



Question. Same circuit, node voltages?

Example 12.4 with node voltages

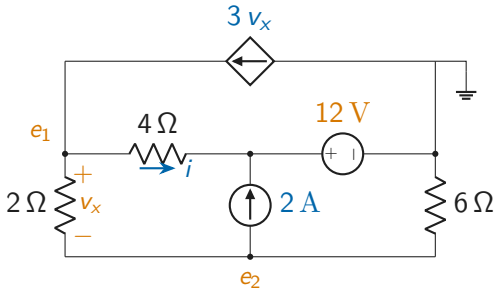


Question. Same circuit, node voltages?

Middle node: 12 V known

Unknowns: e_1, e_2

Example 12.4 with node voltages



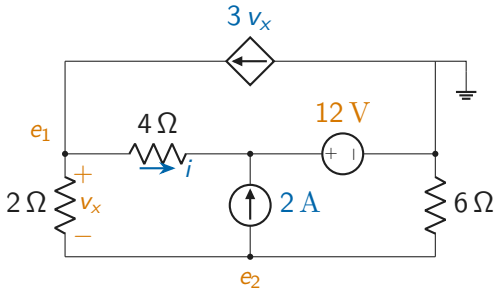
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Node 1: $\dfrac{e_1 - e_2}{2} + \dfrac{e_1 - 12}{4} - 3\,v_x = 0$

Example 12.4 with node voltages



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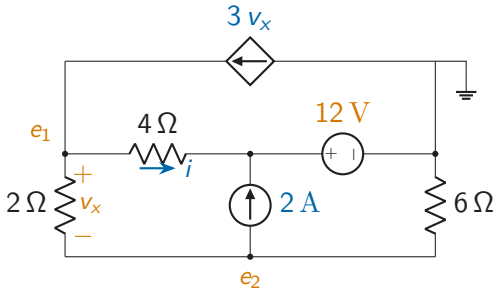
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Node 2: $2 + \dfrac{e_2 - e_1}{2} + \dfrac{e_2}{6} = 0$

Constraint: $v_x = e_1 - e_2$

Example 12.4 with node voltages: matrix form

$v_x = e_1 - e_2$:

$$\begin{bmatrix} -\frac{9}{4} & \frac{5}{2} \\ -\frac{1}{2} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

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$$i = \frac{e_1 - 12}{4} = \frac{-28 - 12}{4} = -10\,\mathrm{A} \quad \checkmark$$

Node or mesh?

Question. Example 12.4: which needs fewer equations?

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Mesh currents

$\ell = 3$, 2 current sources

$3 - 2 = 1$ equation

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$n - 1 = 3$, 1 voltage source

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Mesh currents

$\ell = 3$, 2 current sources

$3 - 2 = 1$ equation

Node voltages

$n - 1 = 3$, 1 voltage source

$3 - 1 = 2$ equations

Tip. Count first. Mesh needs a planar circuit.

Summary

In the circuit	What to do
resistors, voltage sources	$Z_M \mathbf{i}_M = \mathbf{v}_M$, by the rules; symmetric
current source on one mesh	i_{Mk} known; no KVL around mesh k
current source between meshes	Method I or supermesh, plus constraint
dependent source	control in mesh currents: constraint
node or mesh?	count; mesh needs planar

Then $\mathbf{i} = M^T \mathbf{i}_M$ and the element laws